

Roll no.

**Degree: B.Tech. Summer Semester
END-SEMESTER EXAMINATION, June 2026**

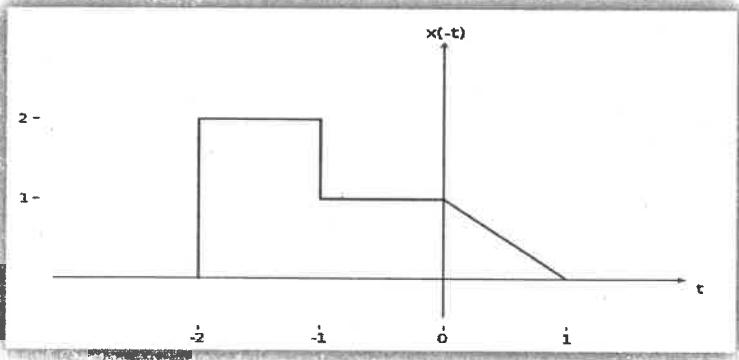
Course Title: Analog and Digital Communication

Course Code: ITECC405 / EIECC405

Duration: 3:00 Hours

Max. Marks: 50

Note: - Attempt any 2(out of 3) parts in all 5 questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q.No.	Questions	Marks	CO
1	(a) For the given signal $x(t)$ compute (i) $x(4t+5)$ (ii) $x(-t)+u(t)$ 	05	CO1
	(b) State and explain the Hilbert Transform with its key properties. Determine the Hilbert Transform of: $x(t) = \text{sinc}(t)$	05	
	(c) Determine the pre-envelope $g_+(t)$ corresponding to the following signal: $g(t) = [1 + k \cos(2\pi f_m t)] \cos(2\pi f_c t)$ where $k < 1$ is a constant.	05	
2	(a) Evaluate the performance of the amplitude modulation (DSB-AM) scheme in the presence of additive white Gaussian noise, assuming a coherent (synchronous) demodulator is used. Derive the expression for the output SNR.	05	CO2
	(b) The carrier $c(t) = A \cos(2\pi \times 10^6 t)$ is angle modulated (FM) by a sinusoidal signal $m(t)$, with the following parameters: $c(t) = A \cos(2\pi \times 10^6 t)$, $m(t) = 4\cos(2000\pi t)$, $k_f = 1500 \text{ Hz/V}$ Determine the frequency deviation Δf and modulation index β.	05	

Q.No.	Questions	Marks	CO
	2 (c) Derive the expression for the canonical (time-domain) representation of an SSB signal obtained by transmitting only the lower sideband (LSB). Sketch the spectrum.	05	
3	(a) Explain the BPSK generation and detection using relevant waveforms.	05	CO3
	(b) State and explain the Sampling Theorem for low-pass signals. Define the Nyquist rate and Nyquist interval.	05	
	(c) Find the Nyquist rates and Nyquist Interval for these signals. i. $x(t) = 7\text{sinc}(40t)\cos(150\pi t)$ ii. $x(t) = 4\text{sinc}(500t) * \cos(200\pi t)$.	05	
4	(a) For the binary data stream 1 0 1 1 1 0 1 0 1, sketch the waveforms for the following line codes: (i) Unipolar NRZ, (ii) Polar NRZ, (iii) Manchester. Compare the three schemes with respect to bandwidth, presence of DC component, and self-synchronization capability.	05	CO4
	(b) Consider the signal set: $s_1(t) = 1, \quad s_2(t) = t/T, \quad 0 \leq t \leq T$ Using Gram-Schmidt orthogonalization, obtain a set of orthonormal basis functions and represent each signal as a vector in the resulting signal space.	05	
	(c) Derive the relationship between Hartley, Nats, Bits	05	
5	(a) A discrete memoryless source has output alphabet $\{a_1, a_2, a_3, a_4, a_5, a_6\}$ with probabilities $\{0.3, 0.15, 0.15, 0.2, 0.1, 0.1\}$. (i) Design a Huffman code for this source. (ii) Determine the average codeword length and the coding efficiency.	05	CO5
	(b) What is Entropy? Derive an equation for maximum Entropy?	05	
	(c) A (6, 3) linear block code has the parity check matrix: $H = [1 \ 0 \ 1 \ 1 \ 0 \ 0; 0 \ 1 \ 1 \ 0 \ 1 \ 0; 1 \ 1 \ 1 \ 0 \ 0 \ 1]$ (rows of H) (i) Find the generator matrix G. (ii) Find the codeword corresponding to the data bits 101.	05	